Turbulence-averaged TF-QKD key rate: result tables and analytical derivation

Content prepared from the repository `Anand-Ambastha/tf-qkd-private`

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1 Scope and conventions

This document collects (i) the tables that show why the log-normal model and the low-order analytic approximations cannot be used at long distance (Sec. 2), (ii) the Gamma–Gamma results with and without beam wander (Sec. 3), (iii) the analytic approximations of the Gamma–Gamma average and their errors, mean versus 2nd-order median versus 4th-order median (Sec. 4), and (iv) the complete derivation from the Gamma–Gamma model to the cumulant–Euler formula and to the log-normal formula (Sec. 5). All formulas and numbers are those of the repository code (`src/`, `analytical_validation/`, `results_v2.3.0/`); the validation scripts were re-run and reproduce the stored result files (70 numerical checks pass, 0 fail).

Provenance. Tables 2, 4, 5, 9, 10 are the repository result tables (`analytical_validation/results/tables/`); Tables 6 and 8 are copied from the repository reports. Tables 3, 7, 11, 12 were computed for this document by calling the repository functions (`legacy_baseline`, `cal_derivatives`, `averaging`, `beam_broadening`, the per-arm Gamma–Gamma channel) at the stated distances; the scripts are in `paper_analysis/scripts/` of the repository.

Beam wander. The repository has no stand-alone beam-wander model for the Gamma–Gamma channel. Beam wander enters through the turbulence-induced *long-term beam broadening* W_{LT} ("GG+BB", `gamma_gamma_bb`), which already contains the wander of the beam centroid; "with beam wander" in this document means GG+BB (Sec. 5.8). Pointing jitter (`corrected_gamma_gamma`) is not used, to avoid counting the same effect twice.

Errors. Every error is 100 (approximation − reference)/reference in percent. References are independent of the formulas under test: Gauss–Hermite/Simpson quadrature in $X = \ln(\eta/\eta_0)$ for the log-normal channel; tensor Gauss–Laguerre quadrature (single Gamma–Gamma channel; checked against Bessel-K quadrature and Monte Carlo); numerical convolution of the Bessel-K log-densities (per-arm Gamma–Gamma).

Table 1: Operating point used in all tables (identical to `analytical_validation/common.py` and `examples/`).

wavelength	$\lambda = 1550$ nm	signal amplitude	$\alpha = 0.$
atmospheric attenuation	0.046 km^{-1}	dark-count probability	$p_d = 1$
transmitter waist	$w_0 = 0.05$ m	phase offsets	$\theta_A = -$
receiver radius	$a = 0.15$ m ($D = 0.30$ m)	photon-number cut-off	$N_{\max}$
C_n^2 classes	10^{-17}, 10^{-15}, 10^{-13} m$^{-2/3}$ (weak / moderate / strong)	rate threshold (max. range)	$R > 1$

2 Why the log-normal model and the low-order approximations fail at long distance

Tables 2–6 give, in this order, the validity distance of the log-normal law, the disagreement between log-normal and Gamma–Gamma key rates, the loss of accuracy of the log-normal analytic formulas as the log-variance grows, the breakdown of the delta method, and the resulting maximum ranges.

Table 2: Validity of the log-normal model. L_{val} is the total distance at which the per-arm plane-wave Rytov variance $\sigma_R^2(L/2) = 1.23 C_n^2 k^{7/6} (L/2)^{11/6}$ reaches 1; $L_{\mathrm{val}} \simeq 0.5547\,\lambda^{7/11}(C_n^2)^{-6/11}$ is the closed form (Eq. 21 of the manuscript). Beyond L_{val} the Rytov perturbation series that underlies the log-normal law no longer converges. Values reproduce the printed Tables I–II of the manuscript.

C_n^2 [m$^{-2/3}$]	L_{val} [km]	closed form [km]	$\sigma_R^2(L/2)$, L=10	L=50	L=100 km
10^{-17}	208.80	208.80	0.003806	0.07277	0.2593
10^{-15}	16.94	16.94	0.3806	7.277	25.93
10^{-13}	1.37	1.37	38.06	727.7	2593

Table 3: Log-normal (LN) versus Gamma–Gamma (GG) per arm at equal per-arm Rytov variance. $\mathrm{SI}_{\mathrm{LN}} = e^{\sigma_R^2} - 1$ grows without bound while $\mathrm{SI}_{\mathrm{GG}}$ saturates near 1. R_{GG}, R_{LN}: turbulence-averaged key rate [bit/pulse]; error $= 100(R_{\mathrm{LN}} - R_{\mathrm{GG}})/R_{\mathrm{GG}}$. Both references are numerical (Bessel-K log-density convolution for GG, Simpson quadrature in X for LN, both with the combined transmittance capped at 1). Rows with $\sigma_R^2 > 1$ of the weak class lie beyond the 252 km deterministic cut-off ($R < 10^{-9}$) and are omitted.

C_n^2	$\sigma_R^2(L/2)$	L [km]	$\mathrm{SI}_{\mathrm{LN}}$	$\mathrm{SI}_{\mathrm{GG}}$	R_{GG}	R_{LN}	LN error [%]
10^{-17}	0.25	98.02	0.28	0.24	4.328×10^{-4}	4.301×10^{-4}	−0.6
10^{-17}	0.50	143.06	0.65	0.43	6.547×10^{-5}	6.367×10^{-5}	−2.8
10^{-17}	1.00	208.80	1.72	0.71	3.914×10^{-6}	3.460×10^{-6}	−11.6
10^{-15}	0.25	7.95	0.28	0.24	3.193×10^{-2}	3.142×10^{-2}	−1.6
10^{-15}	0.50	11.60	0.65	0.43	2.703×10^{-2}	2.566×10^{-2}	−5.0
10^{-15}	1.00	16.94	1.72	0.71	2.163×10^{-2}	1.858×10^{-2}	−14.1
10^{-15}	2.00	24.72	6.39	0.99	1.544×10^{-2}	1.046×10^{-2}	−32.2
10^{-15}	4.00	36.08	53.60	1.17	8.377×10^{-3}	3.568×10^{-3}	−57.4
10^{-13}	0.25	0.64	0.28	0.24	3.710×10^{-2}	3.655×10^{-2}	−1.5
10^{-13}	0.50	0.94	0.65	0.43	3.364×10^{-2}	3.198×10^{-2}	−4.9
10^{-13}	1.00	1.37	1.72	0.71	3.011×10^{-2}	2.579×10^{-2}	−14.4
10^{-13}	2.00	2.00	6.39	0.99	2.720×10^{-2}	1.797×10^{-2}	−33.9
10^{-13}	4.00	2.93	53.60	1.17	2.524×10^{-2}	9.554×10^{-3}	−62.1

Reading the tables.

1. *Validity.* The log-normal model ends at $\sigma_R^2(L/2) = 1$, i.e. at 208.8, 16.9 and 1.37 km for $C_n^2 = 10^{-17}, 10^{-15}, 10^{-13}$ (Table 2). The maximum ranges of Table 6 (≈ 290 km) lie beyond these limits in all three classes.

2. *Model error.* The log-normal scintillation index $e^{\sigma_R^2} - 1$ diverges, while the Gamma–Gamma index saturates (0.71 at $\sigma_R^2 = 1$, 1.17 at 4). Already at $\sigma_R^2 = 1$ the log-normal key rate is 11.6–14.4% below the Gamma–Gamma one, and 57–62% below at $\sigma_R^2 = 4$ (moderate and strong class, Table 3); the log-normal maximum range is 132.7 and 13.1 km against 293.2 and 291.7 km (Table 6).

Table 4: Failure of the log-normal analytic formulas as the log-variance grows (single operating point $L = 80$ km, key rate R; errors against Gauss–Hermite/Simpson quadrature, 100(formula − ref)/ref). "mean is the delta method about $\bar{\eta}$, "2-term and "4-term are Eq. (16) truncated after Θ^2 and Θ^4. γ_1 is the skewness of η. For $\sigma_X^2 \geq 2$ the unclipped reference overflows; the three formulas then give $R = 4.0 \times 10^{-4}$, 2.6×10^{-4}, 3.4×10^{-4} ($\sigma_X^2 = 2$) and 2.9×10^{-4}, 1.2×10^{-4}, 1.8×10^{-4} ($\sigma_X^2 = 3$), i.e. their largest/smallest ratio is 1.5 ($\sigma_X^2 = 2$) and 2.4 ($\sigma_X^2 = 3$).

σ_X^2	$\sigma_R^2(L/2)$	γ_1	R (ref.)	mean [%]	2-term [%]	4-term [%]
0.01	0.02	0.302	1.019×10^{-3}	0.000	−0.002	0.000
0.05	0.10	0.691	9.993×10^{-4}	+0.001	−0.044	−0.001
0.1	0.20	1.01	9.751×10^{-4}	+0.002	−0.173	−0.006
0.307	0.61	2.01	8.811×10^{-4}	+0.026	−1.483	−0.136
0.5	1.00	2.94	8.018×10^{-4}	+0.077	−3.607	−0.513
1	2.00	6.18	6.293×10^{-4}	+0.408	−11.722	−2.963
1.5	3.00	12.1	4.952×10^{-4}	+1.316	−21.793	−7.448
2	4.00	23.7	–	–	–	–
3	6.00	96.5	–	–	–	–

Table 5: Breakdown of the delta method (mean-based expansion) for the key rate. $\gamma_1 = (e^{\sigma_X^2} + 2)\sqrt{e^{\sigma_X^2} - 1}$ is the skewness of η; errors are against quadrature. The last row repeats the expansion in the total transmittance $\eta_{\text{tot}} = \eta^2$. "Appendix H lists the values printed in the manuscript; they are reproduced (the printed "$\sigma_X^2 = 9.6$ for the 140 km case should read 2.4).

L [km]	C_n^2	σ_X^2	γ_1	error [%] (this work)	printed [%]
20	10^{-17}	0.007	0.252	−0.001	−0.001
160	10^{-17}	0.307	2.01	+0.185	+0.180
140	10^{-16}	2.403	41.4	+28.836	+29.000
160 (in η_{tot})	10^{-17}	0.307	2.01	−20.133	−20.100

Table 6: Maximum range (largest total distance with ergodic key rate $> 10^{-9}$ bit/pulse), per-arm models, from `results_v2.3.0/reports/GAMMA_GAMMA_PER_ARM_REPORT.txt`. Every class has its maximum range beyond its own log-normal validity limit, so the log-normal numbers are diagnostic only; Monte Carlo disagrees with the log-normal quadrature at its own maximum-range point by −28% (moderate) and −46% (strong).

C_n^2	L_{val} [km]	deterministic [km]	GG [km]	LN [km]	GG−LN [km]
10^{-17}	208.80	252.04	291.72	317.60	−25.88
10^{-15}	16.94	252.04	293.23	132.73	+160.50
10^{-13}	1.37	252.04	291.72	13.08	+278.64

3. *Analytic formulas inside the log-normal model.* The 4-term formula Eq. (16) has an error of -6×10^{-6}% at $\sigma_X^2 = 0.01$, -0.14% at 0.307, -3.0% at 1.0 and -7.4% at 1.5; the 2-term formula -1.5%, -11.7%, -21.8% at 0.307, 1, 1.5 (Table 4). For $\sigma_X^2 = 2$–3 the largest and smallest of the three formulas differ by a factor 1.5 ($\sigma_X^2 = 2$) and 2.4 ($\sigma_X^2 = 3$).

4. *Delta method.* The skewness of η is $\gamma_1 = 41$ at $\sigma_X^2 = 2.4$ and the mean-based expansion errs by +28.8%; in η_{tot} it errs by -20.1% already at $\sigma_X^2 = 0.31$ (Table 5). Caution: at a given operating point the delta method can be accurate far beyond this (Table 4: within 1.3% up to $\sigma_X^2 = 1.5$ at $L = 80$ km), because $R(\eta)$ is locally close to a power law η^p with $p = \Theta R/R = 1.36, 1.05, 1.13$ at $L = 20, 80, 160$ km (derivatives of the repository), so the curvature term is small. Its accuracy therefore depends on the operating point rather than on a controlled expansion parameter.

3 Gamma–Gamma results with and without beam wander

Per-arm Gamma–Gamma fading (independent arms of length $L/2$, $\eta = \eta_A \eta_B$) without and with turbulence-induced long-term beam broadening, which contains beam wander. Broadening is a deterministic loss on the nominal transmittance of each arm, $\eta_{\text{arm}} \to \eta_{\text{arm}} f_{\text{bb}}$ with $f_{\text{bb}} = [1 - e^{-2a^2/W_{\text{LT}}^2}]/[1 - e^{-2a^2/W^2}]$ and $W_{\text{LT}}^2 = W^2(1 + 1.63\,\sigma_R^{12/5}\Lambda)$ (Sec. 5.8); the fading law and (α, β) are unchanged.

Table 7: Gamma–Gamma (per arm) key rate without (GG) and with (GG+BB) turbulence-induced long-term beam broadening, which contains beam wander ($\alpha = 0.2$ fixed, bit/pulse). W, W_{LT}: free-space and long-term beam radius at one receiver [m]; BB loss $= -10 \log_{10} f_{\text{bb}}^2$ [dB]; R_{det}: deterministic channel without turbulence. Ergodic rates by the repository's 2-D quadrature (300 nodes per arm). †: below the 10^{-9} threshold. Weak-class values agree with an independent exact reference (Table 11 method) to ≤ 0.006%.

C_n^2	L [km]	$\sigma_R^2(L/2)$	W	W_{LT}	BB loss [dB]	R_{det}	R_{GG}	$R_{\text{GG+BB}}$	ratio
10^{-17}	20	0.0136	0.11	0.11	0.00	2.36×10^{-2}	2.36×10^{-2}	2.35×10^{-2}	1.000
10^{-17}	50	0.0728	0.25	0.25	0.08	4.65×10^{-3}	4.57×10^{-3}	4.52×10^{-3}	0.990
10^{-17}	80	0.172	0.40	0.40	0.18	1.02×10^{-3}	9.84×10^{-4}	9.63×10^{-4}	0.978
10^{-17}	100	0.259	0.50	0.50	0.25	4.21×10^{-4}	3.97×10^{-4}	3.85×10^{-4}	0.970
10^{-17}	120	0.362	0.59	0.61	0.32	1.83×10^{-4}	1.69×10^{-4}	1.62×10^{-4}	0.961
10^{-17}	160	0.614	0.79	0.81	0.47	3.72×10^{-5}	3.29×10^{-5}	3.10×10^{-5}	0.941
10^{-15}	20	1.36	0.11	0.15	1.20	2.36×10^{-2}	1.73×10^{-2}	1.51×10^{-2}	0.874
10^{-15}	50	7.28	0.25	0.53	10.72	4.65×10^{-3}	3.68×10^{-3}	9.57×10^{-4}	0.260
10^{-15}	80	17.2	0.40	1.07	16.10	1.02×10^{-3}	7.89×10^{-4}	1.12×10^{-4}	0.142
10^{-15}	100	25.9	0.50	1.50	18.51	4.21×10^{-4}	3.22×10^{-4}	3.25×10^{-5}	0.101
10^{-15}	120	36.2	0.59	1.98	20.45	1.83×10^{-4}	1.40×10^{-4}	9.71×10^{-6}	0.070
10^{-15}	160	61.4	0.79	3.10	23.45	3.72×10^{-5}	2.82×10^{-5}	5.08×10^{-7}	0.018
10^{-13}	20	136	0.11	1.71	36.10	2.36×10^{-2}	1.53×10^{-2}	1.82×10^{-4}	0.012
10^{-13}	50	728	0.25	7.40	55.83	4.65×10^{-3}	3.72×10^{-3}	2.35×10^{-6}	6.3×10^{-4}
10^{-13}	80	$1.72e+03$	0.40	15.69	62.63	1.02×10^{-3}	8.04×10^{-4}	2.22×10^{-9}	2.8×10^{-6}
10^{-13}	100	$2.59e+03$	0.50	22.42	65.42	4.21×10^{-4}	3.28×10^{-4}	$2.41 \times 10^{-17\dagger}$	7.3×10^{-14}
10^{-13}	120	$3.62e+03$	0.59	30.01	67.59	1.83×10^{-4}	1.42×10^{-4}	$0^\dagger$	0
10^{-13}	160	$6.14e+03$	0.79	47.55	70.85	3.72×10^{-5}	2.87×10^{-5}	$0^\dagger$	0

Reading the tables.

1. *Weak class.* Beam wander costs at most 6% of the key rate at $L = 160$ km ($R_{\text{GG+BB}}/R_{\text{GG}} = 0.94$) and 3.8 km (1.3%) of range (Table 8).

2. *Moderate class.* The ratio falls to 0.26 at 50 km, 0.14 at 80 km and 0.018 at 160 km; the range shrinks by 94.6 km (32.3%).

Table 8: Maximum range with and without beam broadening (beam wander), per arm, from `results_v2.3.0/reports/BEAM_BROADENING_REPORT.txt`. Without broadening Gamma–Gamma saturates and the range is nearly independent of C_n^2; with broadening it falls steeply with C_n^2. For $\sigma_R^2 \gg 1$ the broadening law $1.63\,\sigma_R^{12/5}\Lambda$ keeps growing, so the strong-class GG+BB range is a lower bound and plain GG the matching upper bound.

C_n^2	deterministic [km]	det.+BB [km]	GG [km]	GG+BB [km]	BB effect on GG
10^{-17}	252.04	249.06	291.72	287.94	−3.78 km (−1.3%)
10^{-15}	252.04	167.19	293.23	198.61	−94.62 km (−32.3%)
10^{-13}	252.04	63.79	291.72	81.89	−209.84 km (−71.9%)

3. *Strong class.* The long-term beam is 1.7 m at 20 km and 15.7 m at 80 km ($W = 0.11$ and 0.40 m), the rate is below 10^{-9} from ≈ 80 km on, and the range drops by 209.8 km (71.9%). Plain Gamma–Gamma, which only redistributes power around a fixed mean, keeps a range of ≈ 292 km for all three classes.

4. *Assumptions to state when quoting.* Horizontal path, constant C_n^2, collimated Gaussian transmitter; α, β keep their plane-wave values (first-order correction); no extra pointing jitter on top of W_{LT}; the broadening law keeps growing for $\sigma_R^2 \gg 1$, so strong-class GG+BB values are lower bounds.

4 Analytic approximations of the Gamma–Gamma average and their errors

Three approximations are compared, all evaluated with the analytic derivatives of the repository: *mean* (delta method about $\mathbb{E}[\eta]$ or $\mathbb{E}[\eta_{\mathrm{tot}}]$), *2nd-order median* and *4th-order median* (Eq. (11) truncated after Θ^2 resp. Θ^4, expanded about the log-mean point $\eta_m = \eta_0 e^{c_1}$). Table 9 is the progress-report setting (one cumulative channel, aperture-averaged); Tables 11 and 12 use the per-arm model of the code, without and with beam wander.

Table 9: Gamma–Gamma analytic approximations of $\mathbb{E}[R]$, single cumulative channel $\eta_{\mathrm{tot}} = \eta_{\mathrm{FSO}}(L)\,I$, aperture-averaged ($D = 0.30$ m, $\sigma_R^2 = \sigma_R^2(L)$; rows marked * use $L = 80$ km with the prescribed σ_R^2). Errors [%] against tensor Gauss–Laguerre quadrature (confirmed by Bessel-K quadrature to 10^{-8}% and by Monte Carlo): *mean* = delta method about $\mathbb{E}[\eta_{\mathrm{tot}}]$; 2nd, 3rd, 4th = Eq. (11) truncated after Θ^2, Θ^3, Θ^4, expanded about the log-mean ("median) point; "gain is the increase of $\mathbb{E}[R]$ from aperture averaging. The last two columns apply the *log-normal* formula Eq. (16) with the same Rytov variance to the Gamma–Gamma channel. Shape parameters use $(\sigma_R^2)^{6/5}$ (code since v2.2.0); with the report's $(\sigma_R^2)^{12/5}$ the 4th-order error stays ≤ 0.012% (here ≤ 0.023%).

L [km]	σ_R^2	α/β	gain	mean	2nd	3rd	4th	LN 2nd	LN 4th
20	0.05	43.17/161.63	+0.129	−0.002	−0.001	−0.006	0.000	−0.15	−0.13
50	0.26	9.43/19.73	+0.844	−0.080	−0.001	−0.055	+0.002	−1.27	−1.14
80	0.61	5.34/8.26	+1.828	−0.370	+0.021	−0.121	+0.006	−4.07	−3.70
100	0.92	4.50/5.97	+2.532	−0.627	+0.040	−0.171	+0.010	−7.21	−6.54
120	1.29	4.14/4.84	+3.246	−0.865	+0.057	−0.221	+0.014	−11.45	−10.30
160	2.19	4.00/3.88	+4.696	−1.197	+0.084	−0.311	+0.023	−23.38	−20.51
80*	3.00	4.12/6.34	+6.652	−0.634	+0.037	−0.211	+0.014	−32.49	−28.25
80*	5.00	4.58/7.95	+8.457	−0.454	+0.032	−0.154	+0.009	−52.61	−45.99

At $L = 20$ km the progress report prints +0.36% in all formula columns; with three independent references the errors here are $|\cdot| \leq 0.006$% (formula-independent offset in the report, see Sec. 6).

Table 10: Choice of expansion point for the Gamma–Gamma formula (single channel as in Table 9): error [%] of $\mathbb{E}[R]$ when Eq. (11) (2nd and 4th order) is centred at the log-mean ("median), at the mean, or at the mode of η. The log-mean point is best in every row.

L [km]	σ_R^2	log-mean 2nd	log-mean 4th	mean 2nd	mean 4th	mode 2nd	mode 4th
20	0.05	−0.0013	0.0000	+0.0097	+0.0002	−0.0159	−0.0001
50	0.26	−0.0006	+0.0017	+0.1447	+0.0060	−0.2098	−0.0040
80	0.61	+0.0214	+0.0060	+0.4350	+0.0213	−0.6146	−0.0147
100	0.92	+0.0404	+0.0100	+0.6757	+0.0351	−0.9711	−0.0250
120	1.29	+0.0575	+0.0144	+0.9093	+0.0503	−1.3319	−0.0369
160	2.19	+0.0843	+0.0225	+1.3026	+0.0785	−1.9453	−0.0592
80*	3.00	+0.0368	+0.0139	+0.7607	+0.0507	−1.0956	−0.0351
80*	5.00	+0.0318	+0.0090	+0.5563	+0.0313	−0.7730	−0.0206

Table 11: Analytic approximations and their errors for the per-arm Gamma–Gamma channel, **weak class** $C_n^2 = 10^{-17}$, with and without beam wander. Error [%] of $\mathbb{E}[e_X]$, $\mathbb{E}[e_Z]$, $\mathbb{E}[R]$ relative to an exact reference (convolution of the two arms' Bessel-K log-densities, 6001 points; density norm and $\mathbb{E}[I_{\mathrm{tot}}]$ equal 1 to 10^{-11}). *mean*: delta method about $\bar{\eta} = \mathbb{E}[\eta]$ in η-space (as manuscript Tables IV–VI); *2nd*, *4th*: Eq. (11) about $\eta_m = \eta_0 f_{\mathrm{bb}} e^{c_1}$ truncated after Θ^2 and Θ^4, with $c_n = 2^{1-n}\kappa_n(\alpha,\beta)$ at $\sigma_R^2(L/2)$ (plane wave, no aperture). $c_2 = \mathrm{Var}(X)$. Beam wander only shifts η_0 (f_{bb}^2 is the end-to-end factor); the repository does not contain this analytic-with-BB comparison, it is computed here from the repository functions.

L	$\sigma_R^2(L/2)$	c_2	f_{bb}^2	e_X			e_Z			R		
				mean	2nd	4th	mean	2nd	4th	mean	2nd	4th
without beam wander (GG)												
20	0.0136	0.007	–	0.0000	0.0000	0.0000	+0.0001	+0.0002	0.0000	−0.0008	−0.0037	0.0000
50	0.0728	0.036	–	0.0000	0.0000	0.0000	−0.0001	+0.0005	0.0000	−0.0008	−0.024	0.0000
80	0.172	0.084	–	−0.0006	−0.0002	0.0000	−0.0013	+0.0001	0.0000	+0.0008	−0.052	+0.0012
100	0.259	0.123	–	−0.0031	−0.0013	−0.0001	−0.0045	−0.0014	−0.0001	+0.0035	−0.083	+0.0039
120	0.362	0.166	–	−0.013	−0.0056	−0.0003	−0.013	−0.0054	−0.0002	+0.010	−0.132	+0.0099
160	0.614	0.258	–	−0.139	−0.063	−0.0048	−0.082	−0.035	−0.0014	+0.055	−0.300	+0.042
with beam wander (GG+BB, $\eta_0 \to \eta_0 f_{\mathrm{bb}}$)												
20	0.0136	0.007	0.999	0.0000	0.0000	0.0000	+0.0001	+0.0002	0.0000	−0.0008	−0.0037	0.0000
50	0.0728	0.036	0.981	0.0000	0.0000	0.0000	−0.0001	+0.0005	0.0000	−0.0008	−0.024	0.0000
80	0.172	0.084	0.959	−0.0006	−0.0002	0.0000	−0.0013	+0.0001	0.0000	+0.0008	−0.052	+0.0012
100	0.259	0.123	0.944	−0.0032	−0.0013	−0.0001	−0.0046	−0.0015	−0.0001	+0.0036	−0.083	+0.0039
120	0.362	0.166	0.928	−0.013	−0.0059	−0.0003	−0.014	−0.0056	−0.0002	+0.010	−0.131	+0.0100
160	0.614	0.258	0.898	−0.146	−0.066	−0.0050	−0.084	−0.036	−0.0014	+0.057	−0.302	+0.042

Table 12: As Table 11 for the **moderate class** $C_n^2 = 10^{-15}$ ($\sigma_R^2(L/2) = 1.4$–61, $\beta \to 1$, $c_2 = O(1)$). There is no small parameter: the 4th-order formula is no longer better than the 2nd, and with beam wander at $L = 160$ km ($f_{\mathrm{bb}}^2 = 4.5 \times 10^{-3}$, operating point below the key-rate cut-off) every formula fails.

L	$\sigma_R^2(L/2)$	c_2	f_{bb}^2	e_X			e_Z			R		
				mean	2nd	4th	mean	2nd	4th	mean	2nd	4th
without beam wander (GG)												
20	1.36	0.441	–	−0.100	−0.091	−0.102	−2.56	−2.34	−2.65	+11.49	+4.93	+12.56
50	7.28	0.781	–	−0.029	−0.014	−0.011	−0.145	−0.011	−0.122	−0.133	−3.59	+2.06
80	17.2	0.842	–	−0.152	−0.089	−0.033	−0.369	−0.122	−0.054	+0.035	−1.00	+2.20
100	25.9	0.853	–	−0.371	−0.217	−0.077	−0.608	−0.209	−0.049	+0.067	−0.403	+2.23
120	36.2	0.857	–	−0.817	−0.473	−0.162	−0.949	−0.333	−0.061	+0.101	−0.114	+2.30
160	61.4	0.857	–	−3.10	−1.71	−0.481	−2.02	−0.698	−0.064	+0.191	+0.113	+2.62
with beam wander (GG+BB, $\eta_0 \to \eta_0 f_{\mathrm{bb}}$)												
20	1.36	0.441	0.758	−0.055	−0.047	−0.056	−1.32	−1.13	−1.38	+6.75	+1.18	+7.57
50	7.28	0.781	0.085	−0.099	−0.056	−0.019	−0.258	−0.078	−0.043	+0.029	−1.46	+1.60
80	17.2	0.842	0.025	−0.920	−0.528	−0.172	−0.961	−0.345	−0.059	+0.127	−0.383	+2.06
100	25.9	0.853	0.014	−2.67	−1.49	−0.422	−1.79	−0.637	−0.069	+0.215	−0.202	+2.39
120	36.2	0.857	0.009	−5.94	−3.07	−0.585	−2.79	−0.868	+0.097	+0.150	−0.377	+2.74
160	61.4	0.857	0.005	−11.87	−3.70	+1.58	−2.20	+1.56	+2.78	−48.82	−51.31	−41.41

The reference is the unclipped Gamma–Gamma product; the repository engine caps each arm at $\eta \leq 1$, which lowers $\mathbb{E}[R]$ by 9.4% at $L = 20$ km (arm nominal 0.56), $\leq 0.53\%$ at $L = 50$ and $\leq 0.07\%$ for $L \geq 80$ km; the large $L = 20$ km errors of the 4th order are therefore a property of the expansion, not of this cap.

Reading the tables.

1. *Single channel with aperture (Table 9).* For $\sigma_R^2(L)$ between 0.05 and 5 the 4th-order median formula is within 0.023% of quadrature, the 2nd order within 0.084%, the 3rd order within 0.31% and the delta method within 1.2%; the log-normal formula applied to the same channel errs by −1.3% at $\sigma_R^2 = 0.26$ and −52.6% at 5. The 3rd order is worse than the 2nd (Sec. 5.7). The median (log-mean) point beats the mean and the mode in every row (Table 10).

2. *Per arm, weak class, with and without beam wander (Table 11).* For $\sigma_R^2(L/2) \leq 0.61$ the 4th-order median error is $\leq 0.005\%$ for e_X, $\leq 0.0014\%$ for e_Z and $\leq 0.042\%$ for R ($L =$ 160 km); mean and 2nd order give $-0.14\%/-0.066\%$ (e_X), $-0.082\%/-0.036\%$ (e_Z) and $+0.055\%/-0.30\%$ (R). Beam wander changes these errors by less than 0.01 percentage points, because it only shifts $\eta_0 \to \eta_0 f_{\mathrm{bb}}$ ($f_{\mathrm{bb}}^2 \geq 0.90$). For R the delta method is competitive with the 4th order (see the power-law remark in Sec. 2); for e_X and e_Z the 4th order is one to two orders better.

3. *Per arm, moderate class (Table 12).* With $\beta \to 1$ the log-variance c_2 is $O(1)$ and the expansion is no longer controlled: the 4th order errs by +12.6% at 20 km and +2.0 to +2.7% for $L = 50$–160 km, better than the 2nd order only at 50 km and worse for $L \geq 80$ km; with beam wander at 160 km ($f_{\mathrm{bb}}^2 = 4.5 \times 10^{-3}$) the expansion point lies below the key-rate cut-off and all approximations fail (−41% to −51% for R). The formula is therefore reliable for $c_2 \lesssim 0.3$ (weak class and aperture-averaged channels), not in saturated moderate/strong turbulence.

5 Derivation: from the Gamma–Gamma channel to the cumulant–Euler formula

5.1 Channel model and the quantity to be averaged

Let Charlie sit midway, so that each arm has length $L/2$. Writing $\bar{\eta}_A, \bar{\eta}_B$ for the (deterministic) arm transmittances and I_A, I_B for the independent, unit-mean fading factors,

$$\eta_A = \bar{\eta}_A I_A, \qquad \eta_B = \bar{\eta}_B I_B, \qquad \eta \equiv \sqrt{\eta_A \eta_B} = \eta_0\, e^X, \qquad X = \tfrac{1}{2}(\ln I_A + \ln I_B), \qquad \eta_0 = \sqrt{\bar{\eta}_A \bar{\eta}_B}. \tag{1}$$

Here $\eta_0 = \sqrt{\eta_{\rm FSO}(L)}$ for the deterministic link $\eta_{\rm FSO} = \eta_{\rm atm}(L)\,\eta_{\rm geo}(L/2)^2$, and $\eta_0 = \sqrt{\eta_{\rm FSO}}\,f_{\rm bb}$ when turbulence-induced beam broadening (beam wander) is included (Sec. 5.8). The progress-report variant uses one cumulative channel $\eta_{\rm tot} = \eta_{\rm FSO}\,I$, i.e. $X = \frac{1}{2}\ln I$ and $\eta_0 = \sqrt{\eta_{\rm FSO}}$. Every key-rate quantity $Q \in \{p_{XX}, e_X, e_Z, R\}$ is a smooth function $Q(\eta)$ of the effective transmittance, and the turbulence-averaged value is

$$\mathbb{E}[Q] = \int_{-\infty}^{\infty} Q(\eta_0 e^x)\, p_X(x)\, dx. \tag{2}$$

Energy neutrality, $\mathbb{E}[I] = 1$ for each factor, gives $\mathbb{E}[e^{2X}] = 1$, i.e. $\mathbb{E}[\eta_{\rm tot}] = \eta_0^2$.

5.2 Gamma–Gamma statistics and the cumulants of $\ln I$

Product of two Gamma variables. Let $I = X_a X_b$ with $X_a \sim \Gamma(\alpha, \text{rate } \alpha)$ and $X_b \sim \Gamma(\beta, \text{rate } \beta)$ independent, so both have unit mean. For $X \sim \Gamma(k, \text{rate } k)$ the Mellin transform is $\mathbb{E}[X^s] = \Gamma(k+s)/(\Gamma(k)k^s)$ for $s > -k$. By independence

$$\mathbb{E}[I^s] = \frac{\Gamma(\alpha+s)\,\Gamma(\beta+s)}{\Gamma(\alpha)\Gamma(\beta)\,(\alpha\beta)^s}. \tag{3}$$

Mellin inversion gives the familiar density $p_I(I) = \dfrac{2(\alpha\beta)^{(\alpha+\beta)/2}}{\Gamma(\alpha)\Gamma(\beta)} I^{(\alpha+\beta)/2-1} K_{\alpha-\beta}\big(2\sqrt{\alpha\beta I}\big)$ (used in the code as the Bessel-K reference), and (3) gives $\mathbb{E}[I] = 1$, $\mathbb{E}[I^2] = (1+\frac{1}{\alpha})(1+\frac{1}{\beta})$, hence the scintillation index

$$\mathrm{SI} = \frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\alpha\beta}. \tag{4}$$

Cumulants of $Y = \ln I$. Since $\mathbb{E}[e^{sY}] = \mathbb{E}[I^s]$, (3) is the moment generating function of Y, and its logarithm is the cumulant generating function

$$K_Y(s) = \ln\Gamma(\alpha+s) + \ln\Gamma(\beta+s) - \ln\Gamma(\alpha) - \ln\Gamma(\beta) - s\ln(\alpha\beta), \qquad s > -\min(\alpha,\beta). \tag{5}$$

Differentiating at $s = 0$ with $\psi^{(m)}$ the polygamma functions,

$$\boxed{\kappa_1 = \psi(\alpha) + \psi(\beta) - \ln(\alpha\beta), \qquad \kappa_n = \psi^{(n-1)}(\alpha) + \psi^{(n-1)}(\beta) \quad (n \geq 2).} \tag{6}$$

Energy neutrality is the identity $K_Y(1) = \ln\alpha + \ln\beta - \ln(\alpha\beta) = 0$. The signs are fixed by $\psi' > 0$, $\psi'' < 0$, $\psi''' > 0$: $\kappa_2 > 0$, $\kappa_3 < 0$ (left-skewed log-irradiance, deep fades), $\kappa_4 > 0$. For large shapes, $\psi'(k) \simeq 1/k$, $\psi''(k) \simeq -1/k^2$, $\psi'''(k) \simeq 2/k^3$, so κ_3, κ_4 are of higher order in $\varepsilon \sim 1/\alpha + 1/\beta$ than κ_2 (Sec. 5.7). The code checks (6) against direct numerical integration to $\leq 3 \times 10^{-13}$ (test G0).

Shape parameters. As implemented in `averaging.gg_params` and `turbulence/gamma_gamma.py` (plane wave, $\sigma_R^2 = 1.23\,C_n^2 k^{7/6} L^{11/6}$, $d^2 = kD^2/4L$):

$$\alpha = \left[\exp\!\left(\frac{0.49\,\sigma_R^2}{(1+1.11\,\sigma_R^{12/5})^{7/6}}\right) - 1\right]^{-1}, \tag{7}$$

$$\beta = \left[\exp\!\left(\frac{0.51\,\sigma_R^2\,(1+0.69\,\sigma_R^{12/5})^{-5/6}}{(1+0.9d^2+0.62\,d^2\sigma_R^{12/5})^{5/6}}\right) - 1\right]^{-1}. \tag{8}$$

The symbol $\sigma_R^{12/5}$ here means $(\sigma_R^2)^{6/5}$ (σ_R is the Rytov *standard deviation*). The progress report evaluates $(\sigma_R^2)^{12/5}$; the repository code uses $(\sigma_R^2)^{6/5}$ since v2.2.0 (see Sec. 6). The per-arm model of the repository uses $d = 0$ (no aperture averaging) and $\sigma_R^2(L/2)$; the progress-report tables use $\sigma_R^2(L)$ and $D = 0.30$ m. Aperture averaging enlarges β, which keeps the log-cumulants small and is what makes the expansion below accurate in the report's Table 1.

5.3 Expansion in the log-transmittance

Lemma 1 (derivatives with respect to $x = \ln\eta$)**.** *Let* $\Theta = \eta\,\dfrac{d}{d\eta}$ *(Euler operator) and* $F(x) = Q(\eta_0 e^x)$*. Then* $F^{(n)}(x) = (\Theta^n Q)(\eta_0 e^x)$ *for all* $n \geq 0$*.*

Proof. $F'(x) = \eta_0 e^x Q'(\eta_0 e^x) = (\Theta Q)(\eta_0 e^x)$; apply the same step to ΘQ and iterate. □

Lemma 2 (Euler–Stirling form)**.** $\Theta^n = \sum_{k=1}^{n} S(n,k)\,\eta^k \partial_\eta^k$*, with* $S(n,k)$ *the Stirling numbers of the second kind.*

Proof. True for $n = 1$. For the induction step use $\Theta(\eta^k\partial^k f) = \eta\partial(\eta^k\partial^k f) = k\,\eta^k\partial^k f + \eta^{k+1}\partial^{k+1} f$. The coefficient of $\eta^k\partial^k$ in Θ^{n+1} is therefore $kS(n,k) + S(n,k-1) = S(n+1,k)$. □

Explicitly,

$$\Theta^2 = \eta^2\partial^2 + \eta\partial, \qquad \Theta^3 = \eta^3\partial^3 + 3\eta^2\partial^2 + \eta\partial, \tag{9}$$

$$\Theta^4 = \eta^4\partial^4 + 6\eta^3\partial^3 + 7\eta^2\partial^2 + \eta\partial. \tag{10}$$

(The code evaluates $\Theta^n Q$ from the derivatives $Q^{(k)}$, $k \leq 4$, with these weights; `averaging.theta_ops`. Test T2 confirms $\Theta^n\eta^m = m^n\eta^m$ to 10^{-13}.)

Theorem 1 (cumulant–Euler expansion)**.** *Let* c_n *be the cumulants of* $X = \ln(\eta/\eta_0)$*,* $\eta_m = \eta_0 e^{c_1}$ *the log-mean point, and* $\mu_4 = c_4 + 3c_2^2$*. Then*

$$\boxed{\mathbb{E}[Q] \;\simeq\; Q(\eta_m) + \frac{c_2}{2}\,\Theta^2 Q + \frac{c_3}{6}\,\Theta^3 Q + \frac{c_4 + 3c_2^2}{24}\,\Theta^4 Q\Big|_{\eta=\eta_m}, \qquad \eta_m = \eta_0 e^{c_1}.} \tag{11}$$

The first neglected terms are $\frac{\mu_5}{120}\Theta^5 Q + \frac{\mu_6}{720}\Theta^6 Q$ *with* $\mu_5 = c_5 + 10c_2c_3$ *and* $\mu_6 = c_6 + 15c_2c_4 + 10c_3^2 + 15c_2^3$*.*

Proof. Expand F of Lemma 1 about $x = c_1$, $F(x) = \sum_{n=0}^{4} \frac{F^{(n)}(c_1)}{n!}(x-c_1)^n + r_5(x)$, and take the expectation. The central moments $\mu_n = \mathbb{E}[(X-c_1)^n]$ follow from the moment–cumulant relations for the centred variable, whose cumulants are $(0, c_2, c_3, c_4, \dots)$: $\mu_1 = 0$, $\mu_2 = c_2$, $\mu_3 = c_3$, $\mu_4 = c_4 + 3c_2^2$, $\mu_5 = c_5 + 10c_2c_3$, $\mu_6 = c_6 + 15c_2c_4 + 10c_3^2 + 15c_2^3$. With $F^{(n)}(c_1) = (\Theta^n Q)(\eta_m)$ this is (11); the stated neglected terms are the next two terms of the same series. □

Explicit form in ordinary derivatives. Inserting Lemma 2 into (11) and collecting powers,

$$\mathbb{E}[Q] \simeq Q + a_1\,\eta Q' + a_2\,\eta^2 Q'' + a_3\,\eta^3 Q''' + a_4\,\eta^4 Q^{(4)}\Big|_{\eta_m}, \tag{12}$$

$$a_1 = \tfrac{c_2}{2} + \tfrac{c_3}{6} + \tfrac{\mu_4}{24}, \quad a_2 = \tfrac{c_2}{2} + \tfrac{c_3}{2} + \tfrac{7\mu_4}{24}, \quad a_3 = \tfrac{c_3}{6} + \tfrac{\mu_4}{4}, \quad a_4 = \tfrac{\mu_4}{24}. \tag{13}$$

Why the log-mean point. Expanding about $x = c_1 = \mathbb{E}[X]$ removes the first-order term exactly ($\mu_1 = 0$), so the leading correction is the variance term. For a log-normal channel η_m is the median of η; for Gamma–Gamma it is the geometric mean $\exp(\mathbb{E}\ln\eta)$ (called "median" in the report). The alternative expansion points (mean, mode) are compared in Table 10.

5.4 Cumulants of X for the channels used in the paper

Proposition 1. *For the model* (1) *with independent Gamma–Gamma arms of shapes* (α_A, β_A) *and* (α_B, β_B)*,*

$$c_n = 2^{-n}\big(\kappa_n(\alpha_A, \beta_A) + \kappa_n(\alpha_B, \beta_B)\big). \tag{14}$$

For equal arms $c_n = 2^{1-n}\kappa_n(\alpha, \beta)$*; for the single cumulative channel* $\eta_{\mathrm{tot}} = \eta_{\mathrm{FSO}} I$ *one has* $c_n = 2^{-n}\kappa_n(\alpha, \beta)$*.*

Proof. $K_X(s) = \ln \mathbb{E}[e^{sX}] = \ln \mathbb{E}[e^{(s/2)Y_A}] + \ln \mathbb{E}[e^{(s/2)Y_B}] = K_{Y_A}(s/2) + K_{Y_B}(s/2)$ by independence; the n-th derivative at 0 carries the factor 2^{-n}. □

In polygamma form (equal arms):

$$c_1 = \tfrac{1}{2}\kappa_1, \quad c_2 = \tfrac{1}{2}\big[\psi'(\alpha) + \psi'(\beta)\big], \quad c_3 = \tfrac{1}{4}\big[\psi''(\alpha) + \psi''(\beta)\big], \quad c_4 = \tfrac{1}{8}\big[\psi'''(\alpha) + \psi'''(\beta)\big], \tag{15}$$

and for the single channel $c_2 = \frac{1}{4}[\psi'(\alpha) + \psi'(\beta)]$, $c_3 = \frac{1}{8}[\psi''(\alpha) + \psi''(\beta)]$, $c_4 = \frac{1}{16}[\psi'''(\alpha) + \psi'''(\beta)]$. Written in the variable $Y = \ln I$ of the progress report, (11) reads $\mathbb{E}[Q] \simeq Q + \sum_{n=2}^{4} \frac{\mu_n^Y}{n!\,2^n}\Theta^n Q$ with $\mu_2^Y = \kappa_2$, $\mu_3^Y = \kappa_3$, $\mu_4^Y = \kappa_4 + 3\kappa_2^2$, which is exactly `averaging.gg_expansion`. The two forms agree to machine precision in the code (max. relative difference 0 at $L = 50, 100, 160$ km).

5.5 Reduction to the log-normal formula

For a log-normal channel with energy neutrality ($\mathbb{E}[\eta_{\rm tot}] = \eta_0^2$) the log-transmittance is Gaussian, $X \sim \mathcal{N}(-\sigma_X^2, \sigma_X^2)$, so $c_1 = -\sigma_X^2$, $c_2 = \sigma_X^2$, $c_{n\geq 3} = 0$, $\mu_4 = 3\sigma_X^4$. For two arms, $\sigma_X^2 = \sigma_R^2(L/2)/2$. Theorem 1 then gives

$$\mathbb{E}[Q] \simeq Q(\eta_{\rm med}) + \frac{\sigma_X^2}{2}\,\Theta^2 Q + \frac{\sigma_X^4}{8}\,\Theta^4 Q, \qquad \eta_{\rm med} = \eta_0 e^{-\sigma_X^2}, \tag{16}$$

which is the log-normal two-term / four-term formula of the manuscript (the Θ^3 term vanishes with $c_3 = 0$ and $\frac{3\sigma_X^4}{24} = \frac{\sigma_X^4}{8}$). Setting $\kappa_3 = \kappa_4 = 0$ in the Gamma–Gamma formula reproduces (16) to 10^{-12} (test G1). Moments of the log-normal channel: $\mathbb{E}[\eta^n] = \eta_0^n \exp\big[n(n-2)\sigma_X^2/2\big]$, and its skewness in η is $\gamma_1 = (e^{\sigma_X^2} + 2)\sqrt{e^{\sigma_X^2} - 1}$.

5.6 The delta method as the expansion about the mean

Expanding Q about $\bar\eta = \mathbb{E}[\eta]$ in powers of $\eta - \bar\eta$ instead of $\ln\eta - c_1$ gives $\mathbb{E}[Q] \simeq Q(\bar\eta) + \frac{1}{2}Q''(\bar\eta)\,\mathrm{Var}(\eta)$, with $\bar\eta = \eta_0 e^{-\sigma_X^2/2}$, $\mathrm{Var}(\eta) = \eta_0^2(1 - e^{-\sigma_X^2})$ in the log-normal case. The neglected third-order term is $\frac{1}{6}Q'''(\bar\eta)\,\gamma_1 \mathrm{Var}(\eta)^{3/2}$, which is not small once γ_1 is large (Table 5: $\gamma_1 = 41$ at $\sigma_X^2 = 2.4$, error $+28.8\%$). In the variable $x = \ln\eta$ the log-normal law is symmetric, so odd terms vanish and the series is in powers of σ_X^2 only. Carrying out the same delta expansion in the total transmittance $\eta_{\rm tot} = \eta^2$ gives a different and worse result (-20.1% at $L = 160$ km, Table 5).

5.7 Order bookkeeping: which truncation is consistent

For large shapes put $\varepsilon \sim 1/\alpha + 1/\beta$. From (6) and Proposition 1: $c_2 = O(\varepsilon)$, $c_3 = O(\varepsilon^2)$, $c_4 = O(\varepsilon^3)$, hence $\mu_4 = 3c_2^2 + c_4 = O(\varepsilon^2)$, $\mu_5 = O(\varepsilon^3)$, $\mu_6 = O(\varepsilon^3)$. The terms of (11) are therefore

term	$c_2\Theta^2/2$	$c_3\Theta^3/6$	and	$\frac{c_2^2}{8}\Theta^4$
order	ε		ε^2	

and the neglected terms are $O(\varepsilon^3)$. Thus (i) the "2nd-order" formula is correct to $O(\varepsilon)$; (ii) the "4th-order" formula (11) is the complete $O(\varepsilon^2)$ approximation; (iii) the "3rd-order" truncation keeps only one of the two $O(\varepsilon^2)$ terms ($c_3\Theta^3$ but not $\frac{c_2^2}{8}\Theta^4$) and need not improve on the 2nd order. This is the pattern in Table 9: the 3rd-order column is worse than the 2nd-order one, and the 4th-order column is best. The series is asymptotic, not convergent: when no small parameter exists (e.g. $\beta \to 1$ in saturation, $c_2 = O(1)$), adding terms does not help; see the moderate-turbulence rows of Table 12.

5.8 Beam wander / beam broadening

In the repository, turbulence-induced long-term beam broadening (which contains beam wander) acts per arm as a deterministic loss on the nominal transmittance,

$$W_{\mathrm{LT}}^2 = W^2\big(1 + 1.63\,\sigma_R^{12/5}\Lambda\big), \quad \Lambda = \frac{2z}{kW^2}, \qquad f_{\mathrm{bb}} = \frac{1 - e^{-2a^2/W_{\mathrm{LT}}^2}}{1 - e^{-2a^2/W^2}} \le 1, \tag{17}$$

with $z = L/2$, $W = w(z)$ the free-space beam radius, a the receiver radius and $\sigma_R^{12/5} = (\sigma_R^2)^{6/5}$ per arm. The fading statistics are left unchanged (α, β keep their plane-wave values), so in (1) only η_0 changes:

$$\eta_0 \;\longrightarrow\; \eta_0\, f_{\mathrm{bb}}(L/2), \qquad c_n \text{ unchanged.} \tag{18}$$

Theorem 1 therefore applies verbatim with the shifted log-mean point $\eta_m = \eta_0 f_{\mathrm{bb}}\, e^{c_1}$. The derivatives $\Theta^n Q$ must be re-evaluated at the shifted point; they are not rescaled, because $Q(\eta)$ is not a power law. The shift moves the operating point down the key-rate curve towards the cut-off, which is why the expansion deteriorates when f_{bb}^2 becomes small (Table 12, $L = 160$ km).

5.9 Derivatives of the key rate

Equation (11) needs $Q^{(k)}(\eta)$, $k \le 4$, for $Q = e_X, e_Z, R$ (and p_{XX}). With $\gamma = \alpha^2\eta$, $\Omega = \cos\varphi\cos\theta$, $A = 1 - p_d$ the functions are those of the code (`legacy_baseline.py`); all η-derivatives below carry a factor α^{2n} from $\partial_\eta = \alpha^2\partial_\gamma$.

$$p_{XX} = \tfrac{A}{2}\big[e^{-(1-\Omega)\gamma} + e^{-(1+\Omega)\gamma}\big] - A^2 e^{-2\gamma}, \tag{19}$$

$$p_{XX}^{(n)} = \alpha^{2n}\Big\{\tfrac{A}{2}\big[(-(1-\Omega))^n e^{-(1-\Omega)\gamma} + (-(1+\Omega))^n e^{-(1+\Omega)\gamma}\big] - A^2(-2)^n e^{-2\gamma}\Big\}, \tag{20}$$

$$e_X = \frac{\mathcal{N}}{\mathcal{D}}, \quad \mathcal{N} = e^{-\Omega\gamma} - Ae^{-\gamma}, \quad \mathcal{D} = e^{-\Omega\gamma} + e^{\Omega\gamma} - 2Ae^{-\gamma}, \tag{21}$$

$$e_X^{(n)} = \alpha^{2n}\,\frac{\mathcal{N}_\gamma^{(n)} - \sum_{k=1}^{n}\binom{n}{k}\mathcal{D}_\gamma^{(k)}\, e_{X,\gamma}^{(n-k)}}{\mathcal{D}}, \tag{22}$$

where $\mathcal{N}_\gamma^{(n)} = (-\Omega)^n e^{-\Omega\gamma} - A(-1)^n e^{-\gamma}$, $\mathcal{D}_\gamma^{(n)} = (-\Omega)^n e^{-\Omega\gamma} + \Omega^n e^{\Omega\gamma} - 2A(-1)^n e^{-\gamma}$ (derivatives in γ; the recursion is for γ-derivatives, multiplied by α^{2n} at the end). The phase-error bound is

$$p_{ZZ}^{(n_A,n_B)}(\eta) = A\Big[p_d(1-\eta)^N + \sum_{k,l} C_{kl}^{(n_A,n_B)}\,\eta^{k+l}(1-\eta)^{N-k-l} - (1-\eta)^N\Big], \quad N = n_A + n_B, \tag{23}$$

$$S_j = \sum_{\substack{m_A,m_B \\ n_A+n_B\le N_{\max}}} c_{m_A}^{(j)} c_{m_B}^{(j)} \sqrt{p_{ZZ}^{(2m_A+j,\,2m_B+j)}}, \qquad c_m^{(j)} = \frac{e^{-\alpha^2/2}\alpha^n}{\sqrt{n!}}\Big|_{n=2m+j}, \tag{24}$$

$$e_Z = \frac{\sum_{j=0,1}(S_j + D_j)^2}{p_{XX}}, \tag{25}$$

with C_{kl} the coefficients of CAL Eq. (35) and $D_j = \big(\sum_{m\ge 0} c_m^{(j)}\big)^2 - \big(\sum_{m\le m_{\max}} c_m^{(j)}\big)^2$ the truncation residual, which does *not* depend on η (so $D_j^{(n)} = 0$, $n \ge 1$). The key rate is

$$R = 2\,p_{XX}\,K, \qquad K = 1 - h(e_X) - h(e_Z), \qquad h(x) = -x\log_2 x - (1-x)\log_2(1-x). \tag{26}$$

The derivatives are assembled with standard rules:

- $\partial^n[\eta^a(1-\eta)^b] = \sum_{i=0}^{n}\binom{n}{i} a^{\underline{i}}\eta^{a-i}(-1)^{n-i} b^{\underline{n-i}}(1-\eta)^{b-n+i}$ (falling factorials) for p_{ZZ};
- for $\sqrt{p}$, with $p_k = p^{(k)}$ and $s = \sqrt{p_0}$: $(\sqrt{p})' = \frac{p_1}{2s}$, $(\sqrt{p})'' = \frac{p_2}{2s} - \frac{p_1^2}{4p_0^{3/2}}$, $(\sqrt{p})''' = \frac{p_3}{2s} - \frac{3p_1p_2}{4p_0^{3/2}} + \frac{3p_1^3}{8p_0^{5/2}}$, $(\sqrt{p})^{(4)} = \frac{p_4}{2s} - \frac{4p_1p_3+3p_2^2}{4p_0^{3/2}} + \frac{9p_1^2p_2}{4p_0^{5/2}} - \frac{15p_1^4}{16p_0^{7/2}}$;

- quotient $q = u/v$: $q^{(n)} = \big[u^{(n)} - \sum_{k=1}^{n} \binom{n}{k} v^{(k)} q^{(n-k)}\big]/v$ (for $e_Z = U/p_{XX}$);
- entropy: $h' = \log_2 \frac{1-x}{x}$, $h'' = -\frac{1}{x(1-x)\ln 2}$, $h''' = \frac{1}{\ln 2}\big(\frac{1}{x^2} - \frac{1}{(1-x)^2}\big)$, $h^{(4)} = -\frac{2}{\ln 2}\big(\frac{1}{x^3} + \frac{1}{(1-x)^3}\big)$;
- composition (Faà di Bruno, $n \le 4$): $(f\circ g)' = f'g'$, $(f\circ g)'' = f'g'' + f''g'^2$, $(f\circ g)''' = f'g''' + 3f''g'g'' + f'''g'^3$, $(f\circ g)^{(4)} = f'g^{(4)} + f''(4g'g''' + 3g''^2) + 6f'''g'^2g'' + f^{(4)}g'^4$;
- Leibniz for the products $p_{XX}K$ and T_j^2.

These formulas hold on the smooth branch $p_{XX} > 0$, $0 < e_X, e_Z < \frac{1}{2}$, $R > 0$ (no clipping active), which is why an expansion centred below the key-rate cut-off is not meaningful. The analytic derivatives agree with an independent 60-digit `mpmath` implementation to at most 1.0×10^{-9} (all orders 0–4, $L = 20$–160 km; test T1), and the x-space (Hermite) and η-space (Euler–Stirling) routes agree to 3.4×10^{-9} over 78 cross-checks (test T3).

5.10 Summary of the chain

$$\underbrace{I = X_a X_b}_{\text{Gamma–Gamma}} \xrightarrow{(3)} \underbrace{K_Y(s)}_{\text{log-CGF}} \xrightarrow{(6)} \kappa_n \xrightarrow{\text{Prop. 1}} c_n \xrightarrow{\text{Lemmas 1,2, Thm. 1}} \mathbb{E}[Q] \simeq Q(\eta_m) + \tfrac{c_2}{2}\Theta^2 Q + \tfrac{c_3}{6}\Theta^3 Q + \tfrac{c_4+3c_2^2}{24}$$

and, with $\kappa_3 = \kappa_4 = 0$ (Gaussian X), to the log-normal formula (16).

6 Notes on the sources, and reproducibility

Discrepancies found while reproducing the printed numbers.

1. *Gamma–Gamma exponent.* The progress report's Gamma–Gamma figures and tables are reproduced only with $\sigma_R^{12/5}$ evaluated as $(\sigma_R^2)^{12/5}$ (scintillation index peak 0.71, falling to ≈ 0.08 at $\sigma_R^2 = 10$). In Andrews–Phillips σ_R is the Rytov standard deviation, i.e. the exponent is 6/5 on the variance; then SI peaks at ≈ 1.2–1.3 near $\sigma_R^2 = 5$–10 and saturates towards 1. The repository code has used the correct exponent since v2.2.0. The cumulant formulas and the accuracy of the 4th-order formula are unaffected (Table 9), but 26 of the 40 printed Table 1 entries differ by more than 0.25 percentage points under the correct exponent, concentrated in the aperture-gain and delta-method columns at $\sigma_R^2 > 1$.
2. $L = 20$ *km row of the progress report.* It prints +0.36% in the delta, 2nd, 3rd and 4th-order columns; a formula-independent offset points to the reference integral at very large shapes ($\alpha, \beta \approx 43, 162$), not to the expansion.
3. *Manuscript, Eq. (6)/Appendix A.* The A^2 term of p_{XX} sits outside the factor $\frac{1}{2}$: $p_{XX} = \frac{A}{2}[e^{-(1-\Omega)\gamma} + e^{-(1+\Omega)\gamma}] - A^2 e^{-2\gamma}$ (as in CAL Eq. (30) and the code). The printed bracket gives $p_{XX} = 0.5$ instead of 1.57×10^{-3} at $L = 80$ km.
4. *Appendix E.* D_j does not depend on η in CAL and in the code, so $D_j^{(n)} = 0$ for $n \ge 1$; the statement that $D_j^{(n)}$ follows from the $\sqrt{p_{ZZ}}$ terms holds only for a different D_j.
5. *Appendix H.* "$\sigma_X^2 = 9.6$" for $C_n^2 = 10^{-16}$, $L = 140$ km should read 2.4 (it equals $4\sigma_X^2$); the skewness 41 and the error +29% are reproduced.
6. *Naive key rate becoming negative.* In the single cumulative log-normal channel of the report the delta-method key rate first reaches ≤ 0 at $\sigma_R^2 \approx 2.3$ ("near two"); in the two-arm convention of the manuscript it does not become negative up to $\sigma_X^2 = 3$.

Reproducibility. The repository results are regenerated by `python3 analytical_validation/run_all.py` (about 30 s; needs `mpmath`). Tables 7, 3, 11, 12 come from `paper_analysis/scripts/gg_beamwander_table.py`, `ln_vs_gg_per_arm.py` and `perarm_analytic_errors.py` (outputs in `paper_analysis/results/tables/`), which import the repository unchanged.

References

[1] L. C. Andrews and R. L. Phillips, *Laser Beam Propagation through Random Media*, 2nd ed., SPIE Press (2005).

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